



10493 - Weighing the heaviest neutron stars with JWST

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
J0952-0607 g140m				
	1		NIRSpec Bright Object Time Series	(1) PSRJ0952-0607
1311-3430 g140m				
	2		NIRSpec Bright Object Time Series	(2) PSRJ1311-3430
PSR J1653-0158 g140h				
	3		NIRSpec Bright Object Time Series	(3) PSRJ1653-0158
PSR J1810+1744 g140				
	4		NIRSpec Bright Object Time Series	(4) PSRJ1810+1744

ABSTRACT

Black widow pulsars are binaries in which a rapidly spinning neutron star ablates its extremely low-mass companion. Optical observations of several such systems suggest neutron star masses exceeding $2M_{\text{sun}}$ — values that, if confirmed, would rule out many proposed equations of state for ultra-dense matter. However, existing mass estimates are limited by irradiation-driven systematics and low signal-to-noise data from the ground. We propose phase-resolved NIRSpec spectroscopy of the four black widows with the highest inferred neutron star masses. JWST’s infrared sensitivity and stability will enable precise radial-velocity and phase-curve measurements over full orbits, yielding mass constraints a factor 3 tighter than current limits and less vulnerable to systematics. The same data will map temperature contrasts between the day and night hemispheres and probe the dynamics of the ablated wind through the He I $1.083\ \mu\text{m}$ line, providing the first direct constraints on mass-loss rates and shock geometries in black

widow pulsars. These observations will establish the most secure measurements of the maximum neutron star mass and illuminate how pulsar winds strip their companions to produce isolated millisecond pulsars.

OBSERVING DESCRIPTION

Our targets are four compact binaries containing a millisecond pulsar and a low-mass, cold companion. The main goals are to measure radial velocities of the companion over a full orbit and to constrain the atmospheric conditions and mass loss rate of the companions.

We will observe PSRJ0952-0607 and PSRJ1311-3430 (the fainter targets) with the G140M grating, and PSRJ1653-0158 and PSRJ1810+1744 (the brighter targets) with the G140H grating. We will observe the two longer-period targets (PSRJ0952-0607, with $P_{\text{orb}} = 6.42$ hours, and PSRJ1810+1744, with $P_{\text{orb}} = 3.56$ hours) for one orbit each, divided into 50 exposures of duration 7.5 minutes and 4.3 minutes. We will observe the two shorter-period targets (PSRJ1311-3430, with $P_{\text{orb}} = 1.56$ hours, and PSRJ1653-0158, with $P_{\text{orb}} = 1.26$ hours) for two full orbits in order to increase SNR. We will again split each orbit into 50 exposures, with exposure times of 91 seconds and 109 seconds, respectively.

We will acquire all targets using the F110W filter, SUB2048 subarray, and the NRSRAPIDD6 readout pattern, resulting in a total acquisition exposure time of 14.45 seconds. This results in $\text{SNR} = 20\text{-}80$ for all of our targets at phase 0.5-1.0, the window during which we will acquire them.

Proposal 10493 - Targets - Weighing the heaviest neutron stars with JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PSRJ0952-0607	RA: 09 52 8.3214 (148.0346725d) Dec: -06 07 23.49 (-6.12319d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Star Description=[Binary pulsars, Brown dwarfs, Neutron stars, Pulsars] Extended=NO				
	(2)	PSRJ1311-3430	RA: 13 11 45.7213 (197.9405054d) Dec: -34 30 30.38 (-34.50844d) Equinox: J2000	Epoch of Position: 2016	
	Comments: Category=Star Description=[Binary pulsars]				
	(3)	PSRJ1653-0158	RA: 16 53 38.0532 (253.4085550d) Dec: -01 58 36.89 (-1.97691d) Equinox: J2000	Proper Motion RA: -17.3 mas/yr Proper Motion Dec: -3.24 mas/yr Epoch of Position: 2016	
	Comments: Category=Star Description=[Binary pulsars] Extended=NO				
	(4)	PSRJ1810+1744	RA: 18 10 37.2876 (272.6553650d) Dec: +17 44 37.36 (17.74371d) Equinox: J2000	Proper Motion RA: 7.5 mas/yr Proper Motion Dec: -4.189 mas/yr Epoch of Position: 2016	
	Comments: Category=Star Description=[Binary pulsars] Extended=NO				

Proposal 10493 - Observation 1 - Weighing the heaviest neutron stars with JWST

Thu Apr 02 20:00:17 GMT 2026

Observation	Proposal 10493, Observation 1										
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(Observation 1) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	PSRJ0952-0607	RA: 09 52 8.3214 (148.0346725d) Dec: -06 07 23.49 (-6.12319d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Binary pulsars, Brown dwarfs, Neutron stars, Pulsars] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	CLEAR	NRSRAPIDD6	3	1	1	14.452	275194
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSRAPID	500	52	1	1	52	23499.969	275194	
Special Requirements	Time Series Observation No Parallel Attachments										

Proposal 10493 - Observation 2 - Weighing the heaviest neutron stars with JWST

Observation	Proposal 10493, Observation 2										Thu Apr 02 20:00:17 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(Observation 2) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	PSRJ1311-3430	RA: 13 11 45.7213 (197.9405054d) Dec: -34 30 30.38 (-34.50844d) Equinox: J2000			Epoch of Position: 2016					
	Comments: Category=Star Description=[Binary pulsars]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	F140X	NRSRAPIDD6	3	1	1	14.452	275194
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F100LP	NRSRAPID	125	101	1	1	101	11480.92		
Special Requirements	Time Series Observation No Parallel Attachments										

Proposal 10493 - Observation 3 - Weighing the heaviest neutron stars with JWST

Thu Apr 02 20:00:17 GMT 2026

Observation	Proposal 10493, Observation 3										
	Diagnostic Status: Warning										
	Observing Template: NIRSspec Bright Object Time Series										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	PSRJ1653-0158	RA: 16 53 38.0532 (253.4085550d)			Proper Motion RA: -17.3 mas/yr					
			Dec: -01 58 36.89 (-1.97691d)			Proper Motion Dec: -3.24 mas/yr					
			Equinox: J2000			Epoch of Position: 2016					
	Comments: Category=Star Description=[Binary pulsars] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	275194
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140H/F100LP	NRSRAPID	100	101	1	1	101	9203.37	275194	
Special Requirements	Time Series Observation No Parallel Attachments										

Proposal 10493 - Observation 4 - Weighing the heaviest neutron stars with JWST

Observation	Proposal 10493, Observation 4										Thu Apr 02 20:00:17 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSspec Bright Object Time Series										
Diagnostics	(Observation 4) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure.										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	PSRJ1810+1744	RA: 18 10 37.2876 (272.6553650d) Dec: +17 44 37.36 (17.74371d) Equinox: J2000			Proper Motion RA: 7.5 mas/yr Proper Motion Dec: -4.189 mas/yr Epoch of Position: 2016					
	Comments: Category=Star Description=[Binary pulsars] Extended=NO										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	275194
Template	Subarray										
	SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140H/F100LP	NRSRAPID	280	51	1	1	51	12927.606	275194	
Special Requirements	Time Series Observation No Parallel Attachments										